

MGMT 670 Homework 1

Due 11:59 pm EST on September 3. Submit via Brightspace.

1. In smartphone battery testing, 12 phones were tested for battery life (in hours) under two different usage conditions: heavy gaming and normal usage. The following data were recorded:
Heavy Gaming: 4.8, 5.1, 5.3, 4.5, 4.2, 5.0, 4.7, 5.4, 4.9, 5.2, 4.6, 4.4
Normal Usage: 9.5, 10.1, 9.8, 9.2, 9.9, 10.3, 9.7, 10.0, 9.6, 9.4, 10.2, 9.3
 - (a) Calculate the mean battery life under both conditions.
 - (b) Calculate the median of the heavy gaming battery life.
 - (c) Which usage condition has the higher sample standard deviation?
2. Suppose the distribution of weekly grocery spending is bell-shaped with a mean of 150 and a standard deviation of 20. Use the empirical rule to determine the percentage of data within each of the following ranges:
 - (a) 110 to 190
 - (b) 130 to 170
3. Consider the experiment of drawing a single card from a standard 52-card deck.
 - (a) What is the probability of drawing a face card (jack, queen, or king)?
 - (b) If two cards are drawn *without replacement*, what is the probability that at least one of them is a face card?
4. A cybersecurity firm monitors login attempts to detect suspicious activity. From past data, about 3% of all login attempts are from hackers. The system also found that:
 - If attempt is from a hacker, there is a 90% chance it will trigger a security alert.
 - If attempt is from a legitimate user, there is a 5% chance it will (falsely) trigger a security alert.
 - (a) Given that a security alert was triggered, compute the probability that the login attempt was from a hacker.
 - (b) The firm decides to automatically block login attempts if this posterior probability exceeds 50%. Should the system block an account whenever an alert is triggered? Explain.
5. The file Stock_Price_NVDA_MSFT.xlsx contains daily closing prices for technology companies Nvidia (NVDA) and Microsoft (MSFT) in the past year (2024/8-2025/8).
 - (a) Develop a scatter diagram with NVDA stock price on the horizontal axis and MSFT stock price on the vertical axis. What appears to be the relationship between these two stock prices?
 - (b) Compute and interpret sample correlation coefficient. What does this value indicate about the relationship between these two stock prices.